Qatar-10b

R-0147

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-10_230701` · created 2026-07-16T08:40:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460126.8596 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.017
Transit depth (Rp/Rs)^2	1.02 +/- 0.35 [%]
Orbital Inclination (inc)	89.84 +/- 2.69
Airmass coefficient 1 (a1)	1.215 +/- 0.013
Airmass coefficient 2 (a2)	-0.142 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	3.3
Optimal Annulus	12.41
Transit Duration (day)	0.1158 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.101 ± 0.017 vs 0.1265 (deviation 1.43σ)
Duration fit vs published	0.1158 d vs 0.1154 d (deviation 0.09σ)
Mid-time O–C	-15.3 min
Depth SNR	5.6
Residual scatter	1.07 %

Light curve

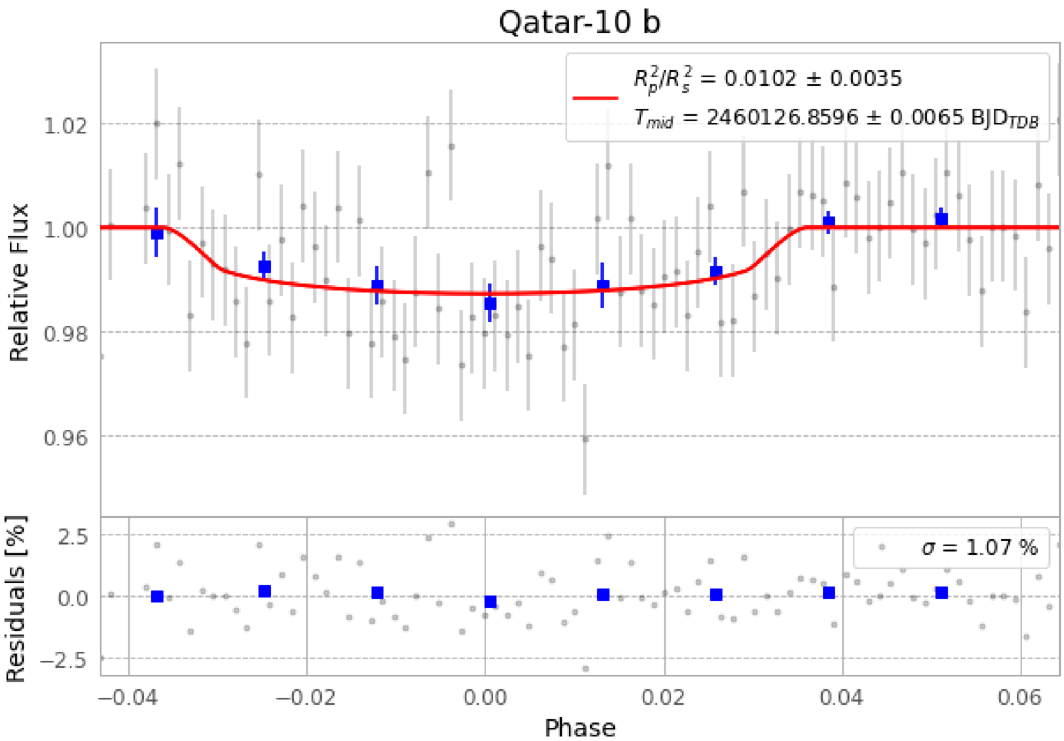

Figure 1 — FinalLightCurve_Qatar-10 b_2023-06-30.png (SHA-256 71d3ccba2fc51922…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 207], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b84ce69c49e4c626…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba`

Data provenance

86 FITS frames (56 MB), `Qatar-10230701065415.FITS` → `Qatar-10230701110915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-230701.log (42f6c8e72b4d…), FinalLightCurve_Qatar-10 b_2023-06-30.png (71d3ccba2fc5…), FinalLightCurve_Qatar-10 b_2023-06-30.csv (523b4b6e4bc5…)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 08:40Z.